

JIE GAO 高洁

Postdoctoral Associate
Mens Manus Machina (M3S)
Singapore-MIT Alliance for Research and Technology
1 Create Way (University Town, NUS),
Singapore 138602

+ 65 88852845
gaojie056@gmail.com
jie_gao@mit.edu
jie.gao@smart.mit.edu
<https://gaojie058.github.io/>

RESEARCH INTERESTS

Human-AI Collaboration, AI-assisted Qualitative Analysis, Human-LLM Interaction

WORK EXPERIENCE

2024.02 - present Singapore-MIT Alliance for Research and Technology, Singapore
Postdoctoral Associate
Advisor: Thomas W. Malone

EDUCATION

2019.09-2024.01 Singapore University of Technology and Design, Singapore
Ph.D. Student in Computer Science
Advisors: Simon Perrault, Kenny Choo
Committee: Wei Lu, Kwan Hui Lim, Darren Herremans

2017.09-2019.06 Zhejiang University, Hangzhou, China
M.Eng. in Software Engineering
Advisors: Cheng Yao and Leijing Zhou

2013.09-2017.06 Shanghai University, Shanghai, China
B.Eng. in Digital Media Technology
Advisors: Feng Tian and Zhifeng Xie

RESEARCH EXPERIENCES

2023.01-2023.07 University of Notre Dame, IN, USA
Visiting Ph.D. Student
Advisor: Toby Jia-jun Li

2022.06-2023.01 National University of Singapore, Singapore
Visiting Ph.D. Student
Advisor: Shengdong Zhao

PUBLICATIONS

Jie Gao, Yuchen Guo, Gionnieve Lim, Tianqin Zhang, Zheng Zhang, Toby Jia-Jun Li, Simon Tangi Perrault. **CollabCoder: A Lower-barrier, Rigorous Workflow for Inductive Collaborative Qualitative Analysis with Large Language Models**. ACM Conference on Human Factors in Computing Systems (CHI2024)

ShunYi Yeo, Gionnieve Lim, Jie Gao, Weiyu Zhang, Simon Tangi Perrault. **Help Me Reflect: Leveraging Self-Reflection Interface Nudges to Enhance Deliberativeness on Online Deliberation Platforms**. ACM Conference on Human Factors in Computing Systems (CHI2024)

Jie Gao, Kenny Tsu Wei Choo, Junming Cao, Roy Ka Wei Lee, Simon Perrault. **CoAICoder: Examining the Effectiveness of AI-assisted Human-to-Human Collaboration in Qualitative Analysis**. ACM Transactions on Computer-Human Interaction (TOCHI2023)

Zheng Zhang, Jie Gao, Ranjodh Singh Dhaliwal, and Toby Jia-Jun Li. **VISAR: A Human-AI Argumentative Writing Assistant with Visual Programming and Rapid Draft Prototyping**. ACM Symposium on User Interface Software and Technology (UIST2023)

Nuwan Janaka, Jie Gao, Lin Zhu, Shengdong Zhao, Lan Lyu, Peisen Xu, Maximilian Nabokow, Silang Wang, Yanch Ong. **GlassMessaging: Towards Ubiquitous Messaging Using OHMDs**. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT2023)

Junming Cao, Bihuan Chen, Longjie Hu, Jie Gao, Kaifeng Huang, Xin Peng. **Understanding the Complexity and Its Impact on Testing in ML-Enabled Systems**. The International Conference on Software Maintenance and Evolution (ICSME2023)

PUBLICATIONS (SHORT PAPER/WORKSHOP/POSTER/PREPRINT)

Jie Gao, Simret Araya Gebreegziabher, Kenny Tsu Wei Choo, Toby Jia-Jun Li, Simon Tangi Perrault, Thomas W. Malone. **A Taxonomy for Human-LLM Interaction Modes: An Initial Exploration**. ACM Conference on Human Factors in Computing Systems (CHI EA 2024)

Jie Gao, Yuchen Guo, Toby Jia-jun Li, Simon Perrault. **CollabCoder: A GPT-Powered Workflow for Collaborative Qualitative Analysis**. ACM Conference On Computer-Supported Cooperative Work And Social Computing (CSCW Demo 2023)

Jie Gao, Xiayin Ying, Junming Cao, Yifan Yang, PinSym Foong, Simon Perrault. **Differences of Challenges of Working from Home (WFH) between China and the US during COVID-19**. ACM Conference on Human Factors in Computing Systems (CHI EA 2022)

Jie Gao, Leijing Zhou, Miaomiao Dong, Fan Zhang. **Expressive Plant: A Multisensory Interactive System for Sensory Training of Children with Autism**. (Ubicomp Poster 2018)

Jie Gao, Pin Sym Foong, Yifan Yang, Weilin Jiang, Yijie Chen, Xiayin Ying, Simon Perrault. **From 996 to 007: Challenges of Working from Home During the Epidemic in China**. arXiv e-prints, arXiv: 2201.09045.

ACADEMIC SERVICES

Associate Chair for CHI Late-breaking Work 2023

Associate Chair for CHI Late-breaking Work 2022

Reviewers for CSCW2022, CSCW2023, CHI2023, AIED2023, CHI2024

TEACHING ASSISTANT

Introduction to Algorithms, Singapore University of Technology and Design, 2020 Fall

- Undergraduate Course
- Instructors: Kai Fong Ernest Chong, Ioannis Panageas

Graphics & Visualization, Singapore University of Technology and Design, 2020 Summer

- Undergraduate Course
- Instructor: Peng Song

MENTORING EXPERIENCE

Ruyuan Wan (PhD Student, University of Notre Dame)

Yuchen Guo (Undergraduate, Singapore University of Technology and Design)

Tianqin Zhang (Undergraduate, Singapore University of Technology and Design)

Shunyi Yeo (PhD Student, Singapore University of Technology and Design)

Yifan Yang (PhD Student, University of Nottingham, Ningbo, China)

TECHNOLOGY SKILLS

Quantitative Data Analysis: Numpy, Pandas, Matplotlib, Seaborn, SciPy, etc.

Qualitative Data Analysis: Grounded Theory, Thematic Analysis, Atlas.ti
Frontend: JavaScript, React.js, Node.js, HTML, CSS
Statistics and NLP: SciPy, scikit-learn, PyTorch, NLTK, etc.
Design Skills: Photoshop, Illustrator, Premiere
Others: MongoDB, Git, LaTeX

[Updated as of May, 2024]